Suponga que $(A \wedge \neg A) \leftrightarrow B$ es falso

determinar el valor de verdad

de $(A \wedge D) \to (B \vee C)$

$$\left.\begin{array}{l} F \leftrightarrow F \\ V \leftrightarrow V \end{array}\right\} V$$

$(A \wedge \neg A) \leftrightarrow B$ $\quad A = V \quad B = V$

$V \wedge F$

$F \leftrightarrow V$

F

$P \to Q$

$V \quad F$

Valor de verdad

$\overbrace{(A \wedge D)}^{P} \to \overbrace{(B \vee C)}^{Q}$ $\quad C = V$ ó F

$V \quad V \qquad V \quad ?$ $\quad D = V$ ó F

$V \to V$

V

$(A \wedge D) \to (B \vee C)$

$V \quad F \qquad V \quad F$

$F \to V$

V

Simplifique

ID $\quad P \to Q$

$\neg P \vee Q$

$(Q \vee R) \vee \neg[(P \to \neg Q) \wedge \neg(P \wedge R)]$

$(Q \vee R) \vee \neg[(\neg P \vee \neg Q) \vee \neg(P \wedge R)]$ ID

$(Q \vee R) \vee [\neg(\neg P \vee \neg Q) \vee \neg\neg(P \wedge R)]$ DM

$(Q \vee R) \vee [(P \wedge Q) \vee (P \wedge R)]$, DM, DN

$(Q \vee R) \vee [P \wedge (Q \vee R)]$, Factor comun

$\underbrace{(Q \vee R)}_{p} \vee [(\underbrace{Q \vee R}_{p}) \wedge P]$, Conmutatividad

$P \wedge (\underbrace{P \vee R}_{p})$

$(Q \vee R)$

$P \rightarrow Q, \neg Q$

$\neg P$

1- $A \wedge B$ ✓

2- $(D \vee C) \vee E$ $\quad \therefore \neg F \wedge A$

3- $F \rightarrow (\neg C \wedge \neg E)$

4- $(B \vee C) \rightarrow \neg D$ $\quad \neg F$

A

5- A Simp 1

6- B ; Simp 1

7. $B \vee C$, Adición 6

8 $\neg D$, mp 7 y 4

9- $\neg(B \vee C) \vee \neg D$, ID 4

10- $\neg D$ $\quad P \rightarrow Q, \neg Q$

$\neg P$

11- $C \vee E \vee D$, Aso 2

12- $((C \vee E) \vee D$, Aso 11

13 $\neg(C \vee E) \rightarrow D$, ID 12

14 $\neg\neg(C \vee E)$, mt 10 y 13

15, $(C \vee E)$ DN , 14

16 $\neg(\neg C \wedge \neg E)$, Dm 15

17 $\neg F$, mt 16 y 3

18 $\neg F \wedge A$, Adi, de 1 y 17

$A = \{\underbrace{2, 3, 4, 6}_{x}\}$ $\quad B = \{\underbrace{2, 3, 4, 5, 6}_{y}\}$

$P(x, y) : x \cdot y > y + 4$

a) $(\exists y \in B)(\forall x \in A)[P(x, y)]$

$y = 5$

$2 \cdot 5 > 5 + 4$ $\quad$ $3 \cdot 5 > 5 + 4$

$10 > 9$ ✓ $\quad$ $15 > 9$

$4 \cdot 5 > 5 + 4$ $\quad$ $6 \cdot 5 > 5 + 4$

$20 > 9$ $\quad$ $30 > 9$

Verdadero $y = 5, 6$ también

$A = \{2, 3, 4, 6\}$ $\quad B = \{2, 3, 4, 5, 6\}$

b) $(\exists y \in N)[P(1, y)]$

$P(1, y) : 1 \cdot y > y + 4$

$1 \cdot 100 > 100 + 4$

$100 > 104$

Falso, just $y = 100$

1 x cualquier numero es cualquier numero y la desigualdad $1 \cdot y > y + 4$ es Falso

$A = \{5, 9, 12, 18, 17, 19\}$ $C = \{11, 18, 17, 16\}$

$B = \{4, 7, 12, 17, 16, 19\}$

a) Los elementos de $P[(A \cap B) - C] \times (C - B)$

$A \cap B = \{12, 17, 19\}$ $C - B = \{11, 14\}$

$A \cap B - C = \{12, 19\}$

$P[(A \cap B) - C]$

$P[12, 19]$

$P[(A \cap B) - C] = \{\emptyset, \{12\}, \{19\}, \{12, 19\}\}$

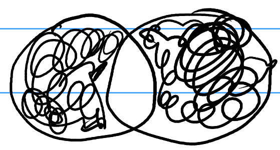
$P[(A \cap B - C] \times (C - B)$

$\{\{\emptyset, \{12\}, \{19\}, \{12, 19\}\}\} \times \{\{11, 14\}\}$

$= \{(\emptyset, 11), (\emptyset, 14), (\{12\}, 11), (\{12\}, 14),$
$(\{19\}, 11), (\{19\}, 14),$
$(\{12, 19\}, 11), (\{12, 19\}, 14)\}$

$A = 1, 2, 3$
$B = 3, 4$

$A \Delta B = A \cup B - A \cap B$



$A \cup B = 1, 2, 3, 4$
$A \cap B = 3$

$A \Delta B = 1, 2, 4$

Elementos → Conjuntos

Valor(es) → Cardinalidad

$A = \{5, 9, 12, 14, 17, 19\}$ $\quad$ $C = \{11, 14, 17, 18)$

$B = (4, 7, 12, 17, 18, 19\}$

B) El valor de $|P[(A \cup B)] \times P(C)|$

$|A| = 6$ $\quad$ $|C| = 4$ $\quad$ $A \cup B = \{5, 9, 12, 14, 17, 19, 4, 7, 18\}$

$|B| = 6$

$|A \cup B| = 9$

$P(A \cup B) = 2^{9} = 512$ $\qquad$ $P(C) = 2^{4} = 16$

Cardinalidad es $2^{|A \cup B|} \wedge 2^{|C|}$

$|P[(A \cup B)] \times P(C)|$

$512 \cdot 16$

$\boxed{8192}$

$A \times (B-C) \subseteq (A \times B) \cap (A \times \bar{C})$

Analizar final

$(x,y) \in (A \times B) \cap (A \times \bar{C})$

$(x,y) \in (A \times B) \wedge (x,y) \in (A \times \bar{C})$

$\underline{x \in A \wedge} \; y \in B \wedge \underline{x \in A \wedge} \; y \in \bar{C}$

$x \in A \wedge y \in B \wedge y \in \bar{C}$, Idem.

Demostracion

$p \wedge p = p$
$p = p \wedge p$
$p = p \vee p$
$p \vee p = p$

$(x,y) \in A \times (B-C)$

$x \in A \wedge y \in (B-C)$

$x \in A \wedge y \in B \wedge y \notin C$

$y \in \bar{C}$
$y \notin C$

$x \in A \wedge y \in B \wedge x \in A \wedge y \notin C$ Idemp.

$(x,y) \in (A \times B) \wedge (x,y) \in (A \times \bar{C})$

$(x,y) \in (A \times B) \cap (A \times \bar{C})$

Def $(x,y) \in (A \times B)$

$x \in A \wedge y \in B$

$ACq \to ACq$

$[C\bar{B} \subseteq C \wedge (D \cup A) \subseteq \bar{C}] \to (D \cap E) \subseteq B$

P_1 - $x \in \bar{B} \to x \in C$ $\qquad x \in B$

$x \notin B \to x \in C$, $x \notin C \to x \in B$

P_2 - $x \in (D \cup A) \to x \in \bar{C}$ $\qquad P \to Q$, $\neg Q$

$x \in D \vee x \in A \to x \notin C$ $\qquad \neg P$

Demo $\qquad P$

$\qquad P \vee ?$

$x \in D \cap E$ $\qquad$ Adición

$x \in D \wedge x \in E$

$x \in D$, Simp

$x \in D \vee x \in A$, Adición

$x \notin C$, P_2

$x \in B$, P_1

$U = \{a, b, c, d, e, f\}$

$A = \{a, b, e\}$ $\quad B = \{c, e, f\}$ $\quad C = \{b, e, f\}$

$\{(A - B) \cup C\} \cap \overline{\underbrace{(B \Delta C) \cup \{d\}}_{\star}}$ $\qquad U - \star$

$A - B = \{a, b\}$

$(A - B) \cup C = \{a, b, e, f\}$ $\qquad A \cup B - A \cap B$

$B \Delta C = B \cup C - B \cap C$

$B \cup C = \{c, e, f, b\} - B \cap C = \{e, f\}$

$B \Delta C = \{c, b\}$

$(B \Delta C) \cup \{d\} = \{c, b, d\}$

$\overline{(B \Delta C) \cup \{d\}} = \{a, e, f\}$

$\{(A - B) \cup C\} \cap \{\overline{(B \Delta C) \cup \{d\}}\}$

$\{a, b, e, f\} \cap \{a, e, f\}$

$\{a, e, f\}$ //